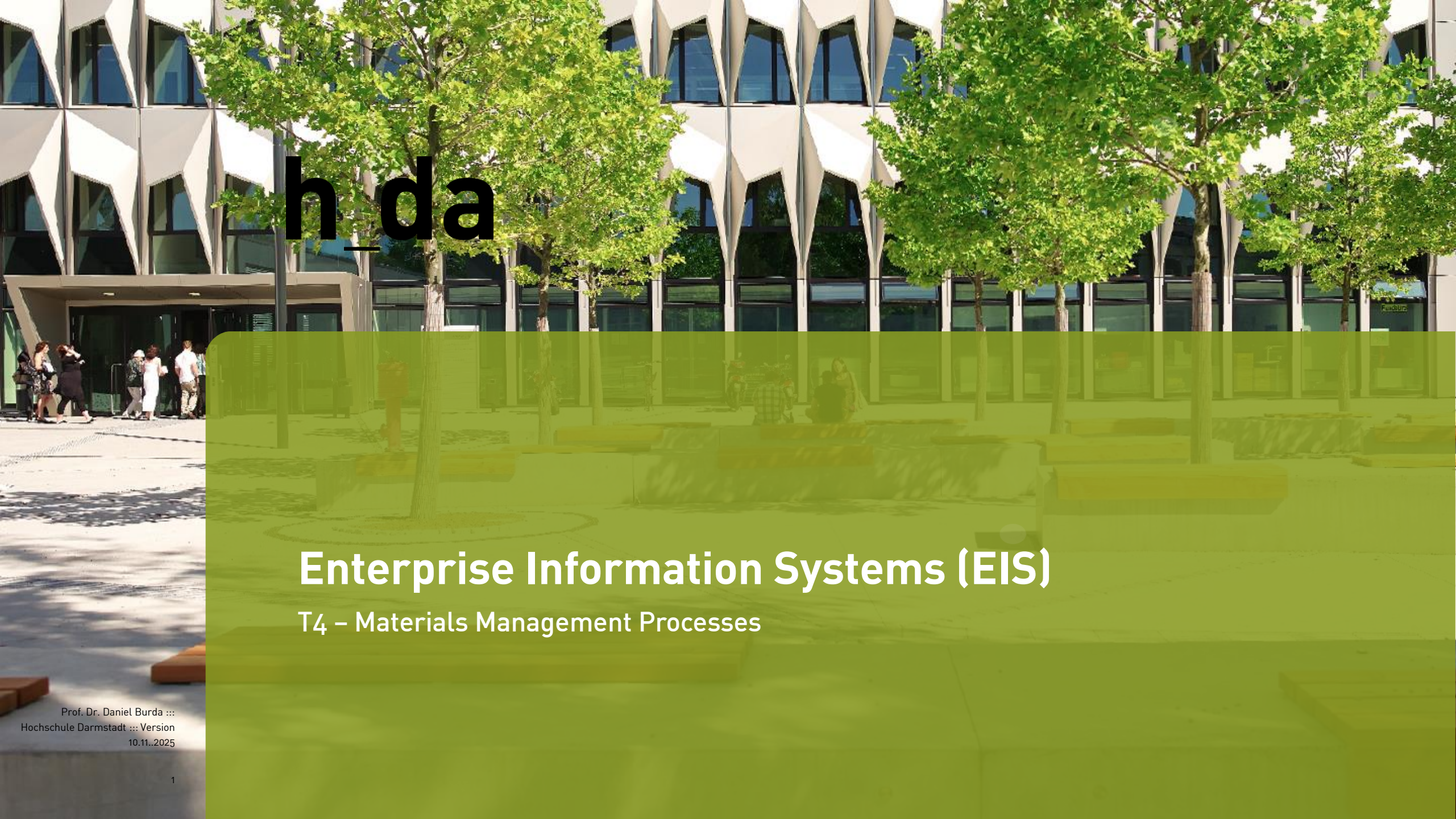


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Enterprise Information Systems (EIS)

T4 – Materials Management Processes

Prof. Dr. Daniel Burda :::
Hochschule Darmstadt ::: Version
10.11..2025

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Schedule (When?)

Lecture				Practical			
No.	Date/Time/Room	Topic No.	Topic	PVL Relevant	Topic	Practical No.	Dates (2 Groups)
1	Tue., 07.10.2025, 10:15 - 11:45 D14 / 04.04	T0	Intoduction & Module Overview	---	---	---	---
2	Tue., 14.10.2025, 10:15 - 11:45 D14 / 04.04	T1	Enterprise Information Systems - Theoretical Perspective	---	---	---	---
3	Tue., 21.10.2025, 10:15 - 11:45 D14 / 04.04	T2	SAP Overview and Introduction S/4 HANA	---			
4	Tue., 28.10.2025, 10:15 - 11:45 D14 / 04.04	T3	Sales and Distribution Processes (1)	---	SAP Navigation Exercises	P#1	28.10.2025 / 29.10.2025
5	Tue., 04.11.2025, 10:15 - 11:45 D14 / 04.04	T3	Sales and Distribution Processes (2)	x	SD Case Study	P#2	04.11.2025 / 05.11.2025
6	Tue., 11.11.2025, 10:15 - 11:45 D14 / 04.04	T4	Materials Management and Procurement Processes (1)	x	SD Case Study	P#2	11.11.2025 / 12.11.2025
7	Tue., 18.11.2025, 10:15 - 11:45 D14 / 04.04	T4	Materials Management and Procurement Processes (2)	x	MM Case Study	P#3	18.11.2025 / 19.11.2025
8	Tue., 25.11.2025, 10:15 - 11:45 D14 / 04.04	T5	Financial Accounting Processes (1)	x	MM Case Study	P#3	25.11.2025 / 26.11.2025
9	Tue., 02.12.2025, 10:15 - 11:45 D14 / 04.04	T5	Financial Accounting Processes (2)	x	FI Case Study	P#4	02.12.2025 / 03.12.2025
10	Tue., 09.12.2025, 10:15 - 11:45 D14 / 04.04	T6	Integrated Project Management (1)	x	FI Case Study	P#4	09.12.2025 / 10.12.2025
11	Tue., 16.12.2025, 10:15 - 11:45 D14 / 04.04	T6	Integrated Project Management (2)	x	PS Case Study	P#5	16.12.2025 / 17.12.2025
12	Tue., 13.01.2026, 10:15 - 11:45 D14 / 04.04	T7	Human Capital Management Processes (1)	x	PS Case Study	P#5	13.01.2026 / 14.01.2026
13	Tue., 20.01.2026, 10:15 - 11:45 D14 / 04.04	T7	Human Capital Management Processes (2)	x	HCM Case Study	P#6	20.01.2026 / 21.01.2026
14	Tue., 27.01.2026, 10:15 - 11:45 D14 / 04.04	n/a	Test Exam / Q&A Final Exam	x	HCM Case Study	P#6	27.01.2026 / 28.01.2026

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Learning Objectives (What?)

Students can…

- … describe the **general Procure-to-Pay Process on the basis of the essential activities** and **differentiate it from an Order-to-Cash process**.
- … describe the **organizational structures and master data** required for Materials Management with SAP MM.
- … create **vendor and material master data relevant for Materials Management** with SAP-MM.
- … create purchase requisition, purchase order and post a goods receipt in SAP-MM.
- … **explain the integrated order-to-cash process and execute** it in SAP-MM.

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Agenda

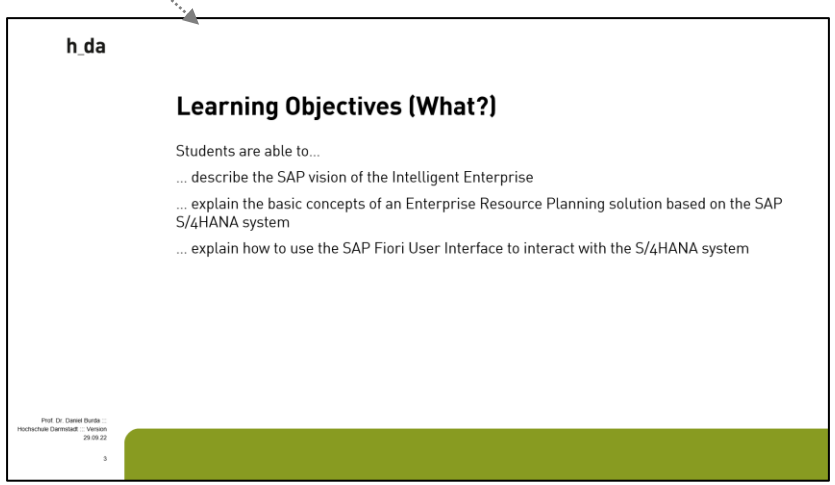
- Purchase-to-pay vs. Order-to-cash processes
- MM Organizational Structure
- MM Master Data
- Procure-to-Pay Process
- Outlook Case Study MM

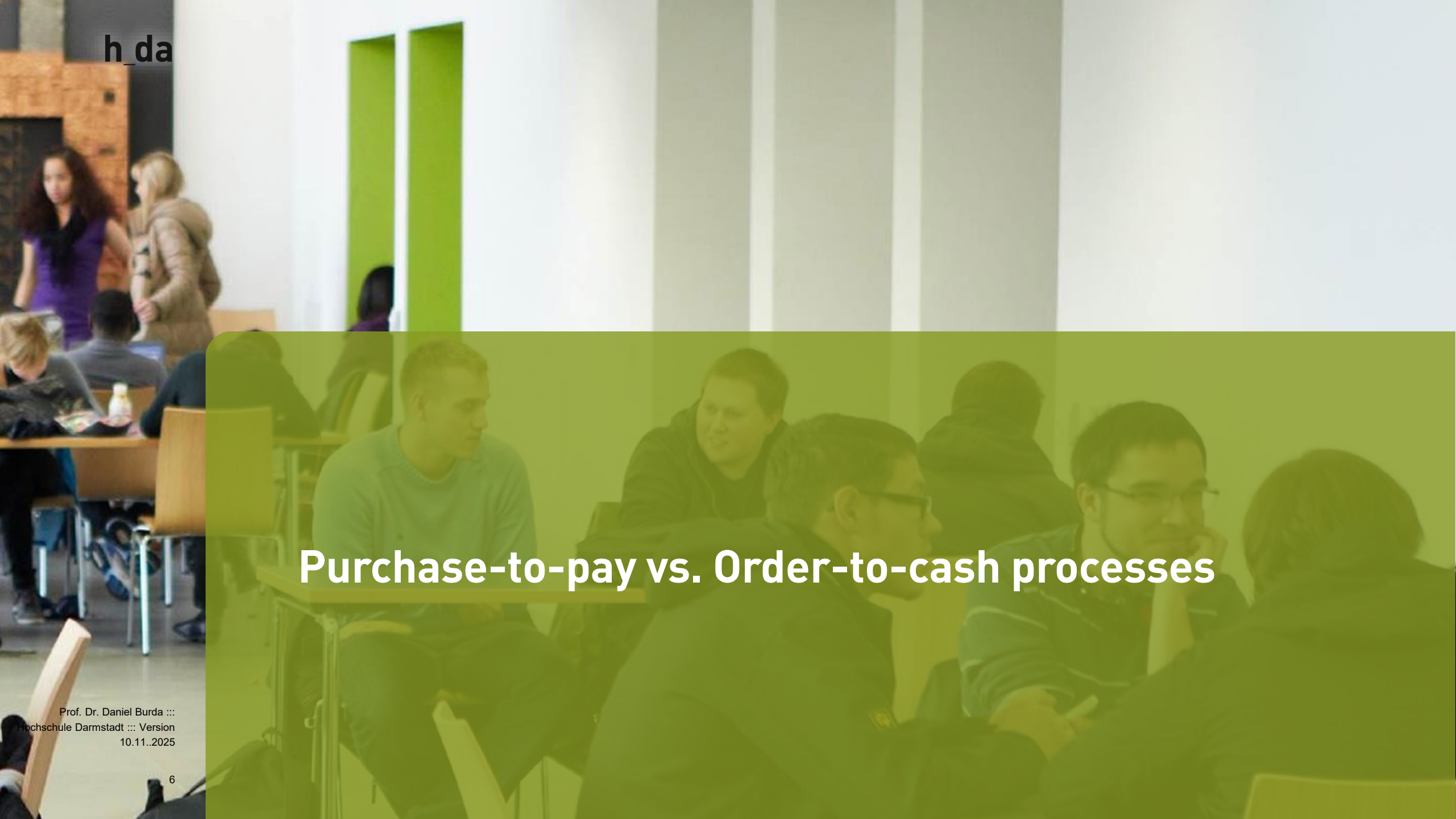
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Purchase-to-pay vs. Order-to-cash processes

Repetition

Purchase-to-pay vs. Order-to-cash

- "Purchase-to-pay/procure-to-pay" processes refer to all activities from the creation of an procurement order to payment of the due amounts to the supplier.
- This is the customer's view of a purchase order process and is therefore associated with the **procurement function** of the company.

- "Order-to-cash" processes refer to all activities from the customer order to the receipt of payment.
- It is the supplier's view of an order process and is therefore associated with of the **sales function** in a company.

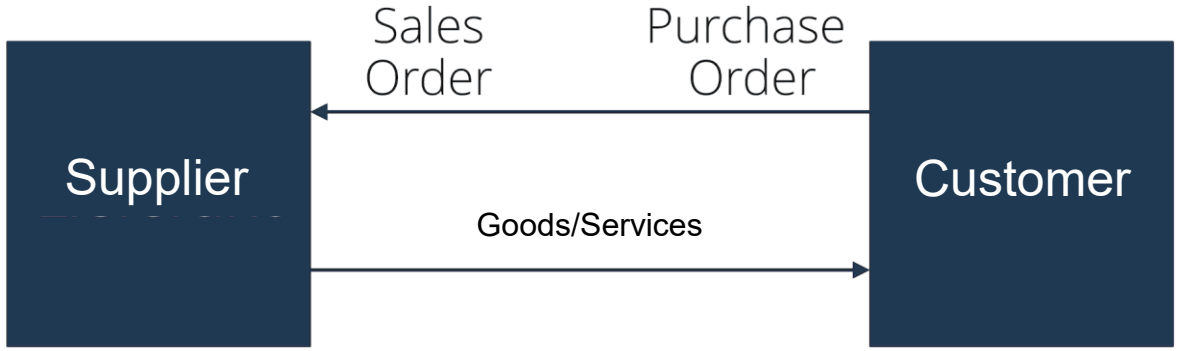


https://blog.esker.com/wp-content/uploads/2020/10/CCC_Petales_Cycle-US-1020x416.png

Repetition

Purchase Order vs. Sales Order

- A purchase order is usually created/transmitted in the form of a document by a customer with a supplier.
- A purchase order sent by the customer becomes a sales order on the supplier side.
- A **purchase order is associated with the procurement** function in a company and is used in a purchase-to-pay process.
- A **sales order is associated with the sales function** of a company and is used in an order-to-cash process.



Quelle: https://ecosio.com/de/blog/unterschied-zwischen-einer-purchase-order-und-einer-sales-order/

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Assignment

Assignment 1: Purchase-to-pay Process

Tasks:

- Based on your understanding, model an Purchase-to-pay process starting from the initial demand to the payment of the vendor´s invoice.
- Concentrate on the essential process steps and on the standard case ("happy path") without exceptions, complications, etc.
- Present your process in the plenum.

Notes:

Choose a graphical notation of your choice for modeling, e.g. a simple flow chart or BPMN, EPK, UML.

You can use a modeling tool of your choice, e.g.: https://online.visual-paradigm.com/app/diagrams/#diagram:proj=0&type=Flowchart

Materials Management (MM)

Curriculum: Introduction to S/4HANA using Global Bike



SAP University Alliances

SAP UCC Magdeburg

Teaching material - Information

Teaching material - Version

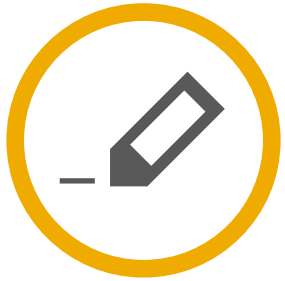
- 4.2 (August 2023)

- Software used
  - S/4HANA 2022
  - Fiori 3.0
- Model
  - Global Bike

- Prerequisites
  - No Prerequisites needed

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Module Information



Authors

- Bret Wagner
- Stefan Weidner
- Babett Ruß



Target Audience

- Beginner

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Unit Overview

- MM Organizational Structure
- MM Master Data
- Procure-to-Pay Process

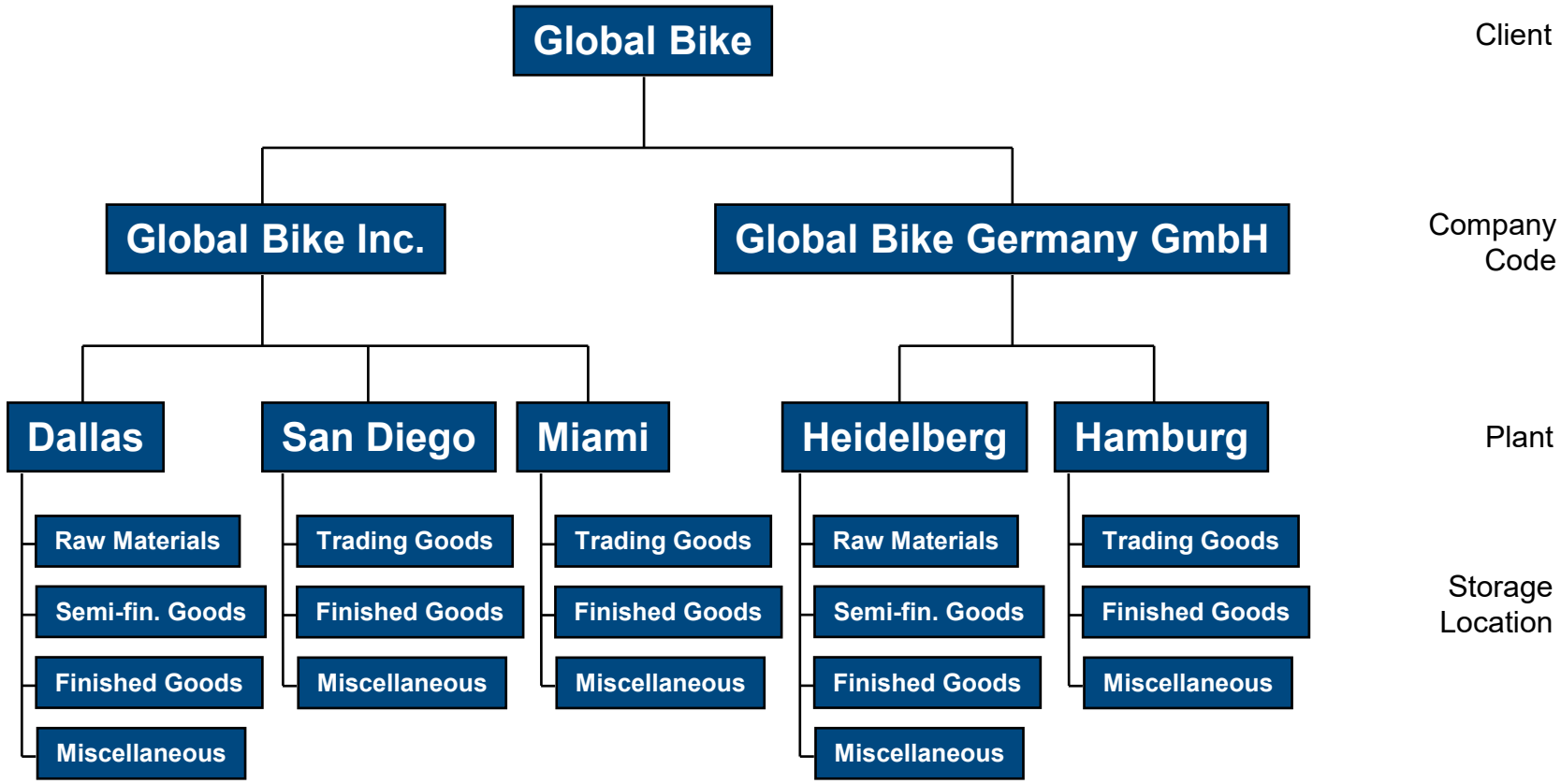
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MM Organizational Structure (Materials Mgmt.)

- Client
  - An independent environment in the system
- Company Code
  - Smallest org unit for which you can maintain a legal set of books
- Plant
  - Operating area or branch within a company
    - Manufacturing, distribution, purchasing or maintenance facility
- Storage Location
  - An organizational unit allowing differentiation between the various stocks of a material in a plant
- Purchasing Organization
  - The buying activity for a plant takes place at the purchasing organization
  - Organization unit responsible for procuring services and materials
  - Negotiates conditions of the purchase with the vendors
- Purchasing Group
  - Key that represents the buyer or group of buyers who are responsible for certain purchasing activities
  - Channel of communication for vendors

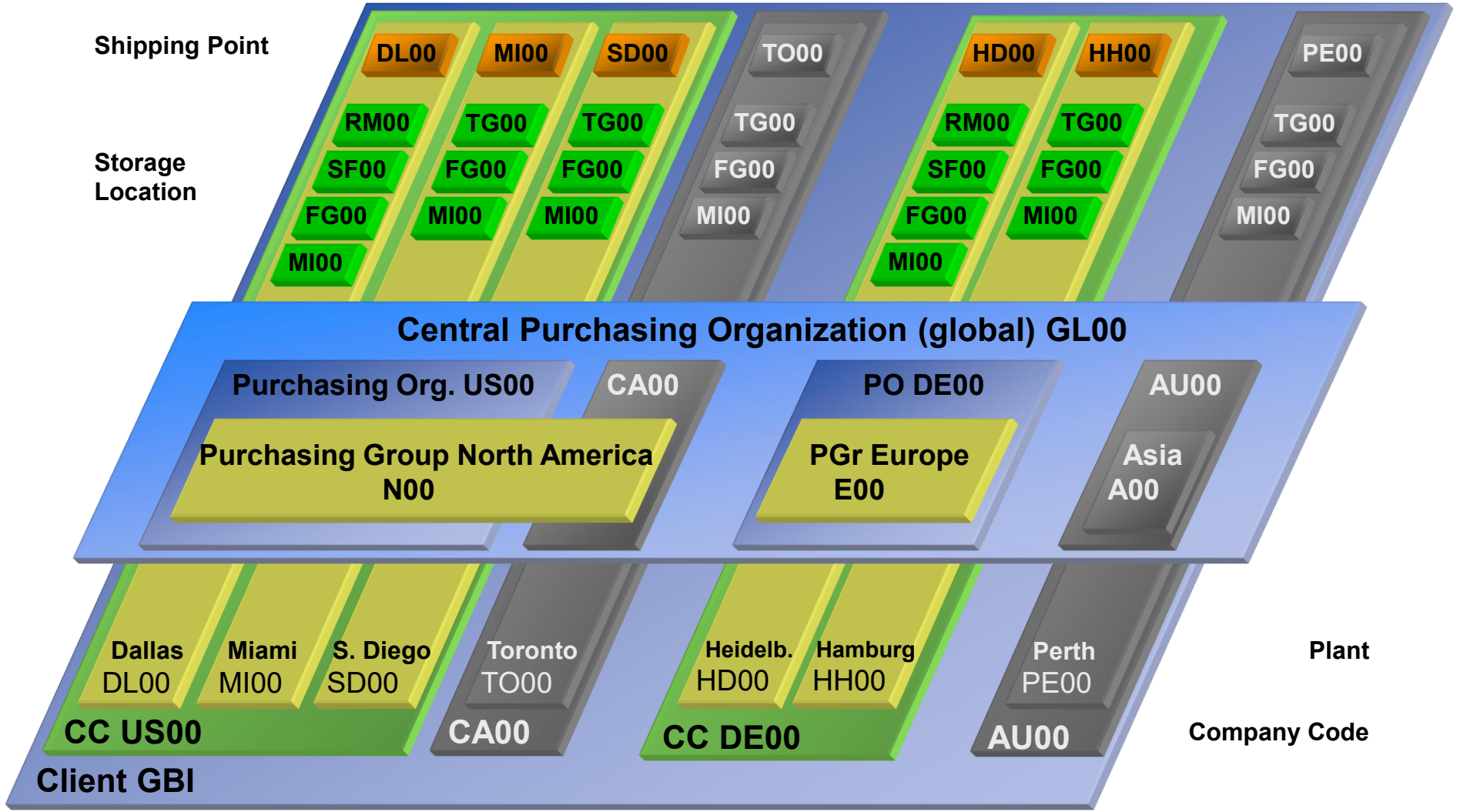
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Global Bike Structure for Materials Management



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Global Bike Enterprise Structure in SAP ERP (Logistics)



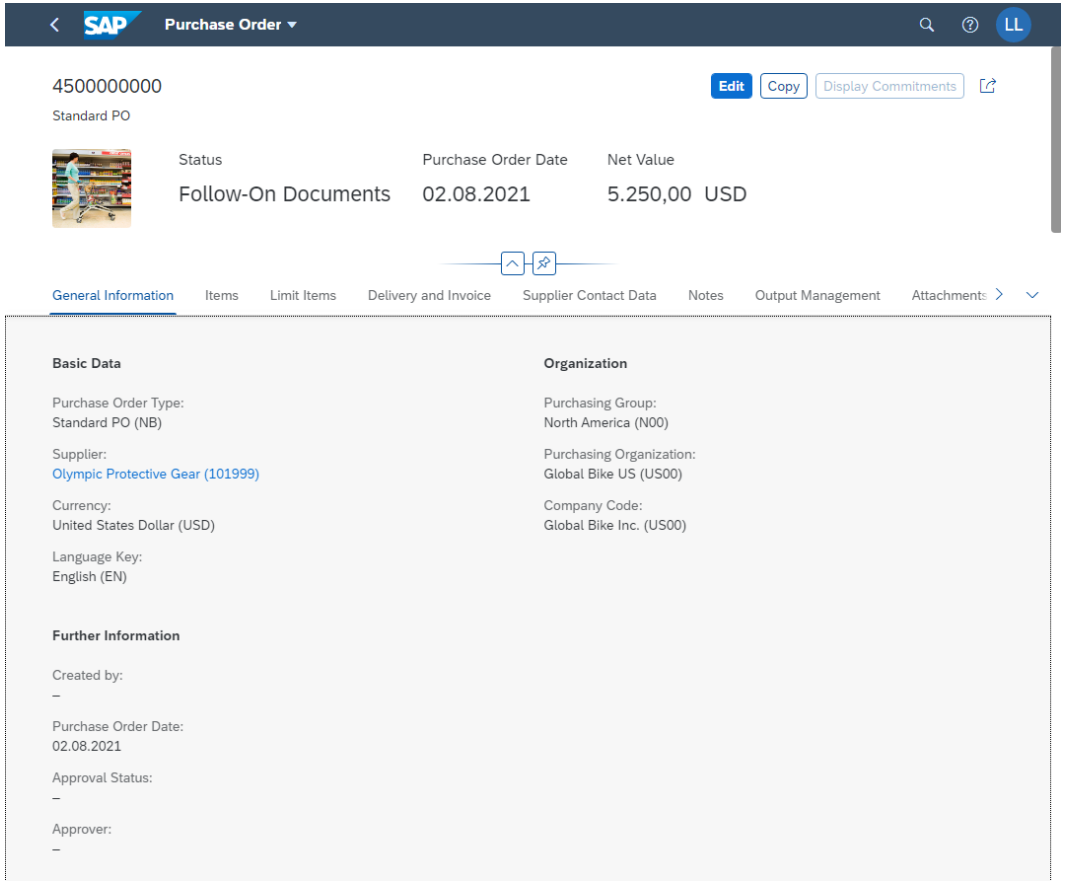
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Unit Overview

- MM Organizational Structure
- MM Master Data
- Procure-to-Pay Process

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MM Master Data

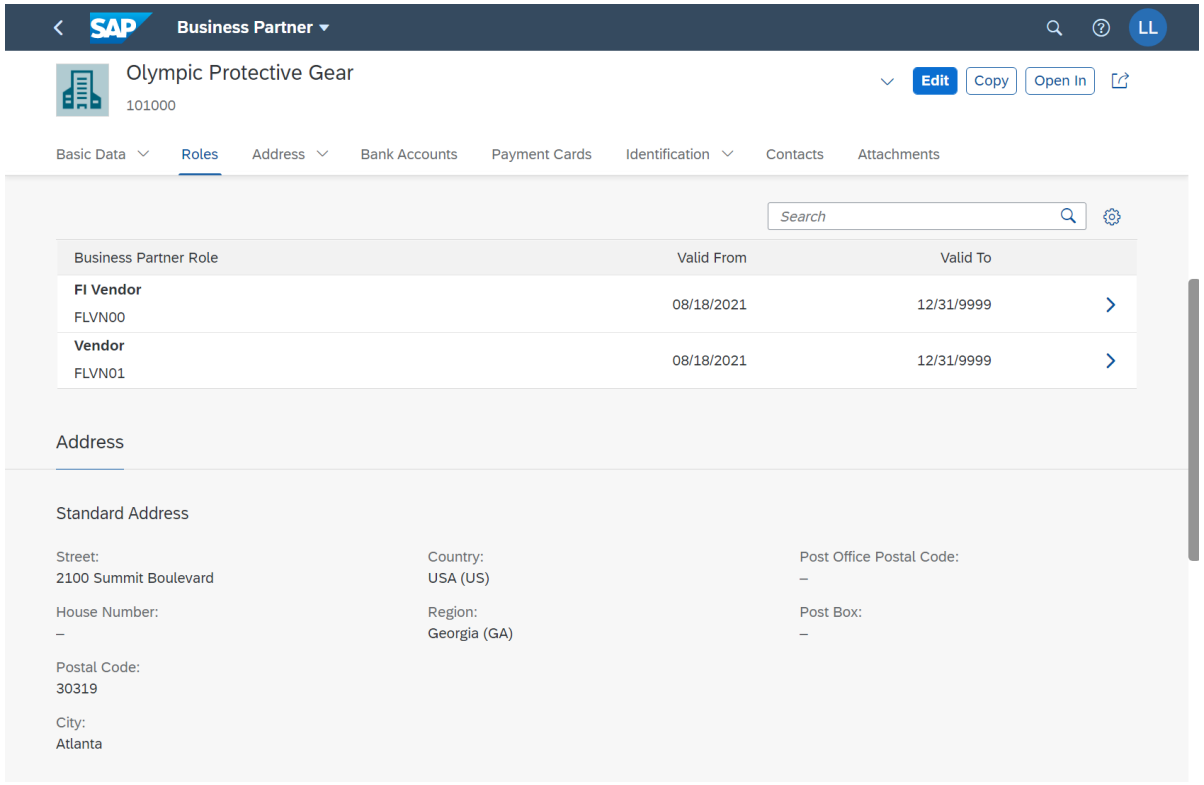


- Vendor Master Data
- Material Master Data
- Purchasing Info Record
- Condition Master Data
- Output Master Data

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Vendor Master Data

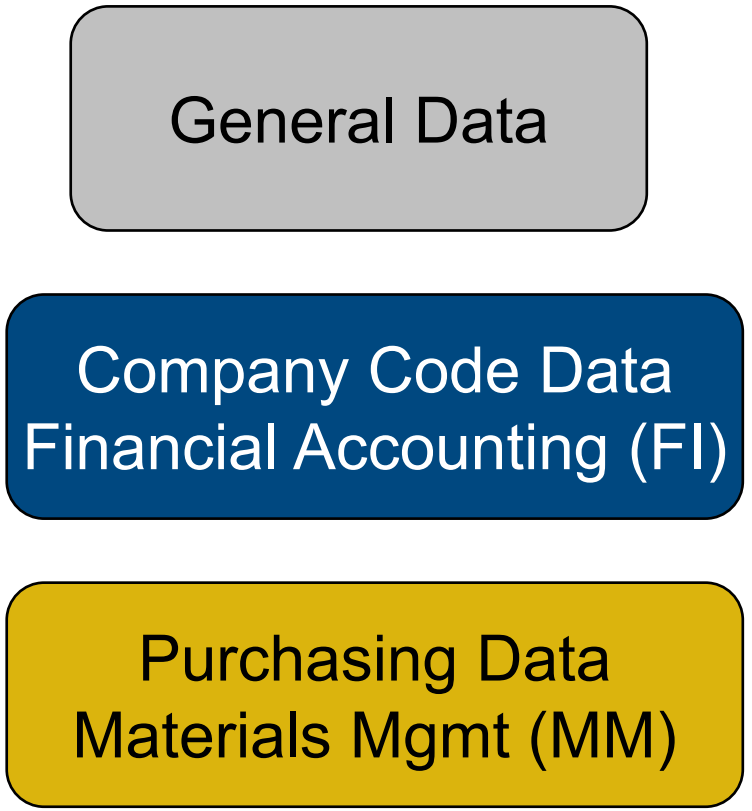
- Vendor Master
  - Contains all the necessary information **needed to do business** with an external supplier
  - Used and maintained primarily by the Purchasing and Accounting Departments
  - Every **vendor MUST have a master** record
  - The supplier master record is created in SAP S/4HANA as a central **business partner master record with different roles**.
  - For suppliers, these are often the business partner roles:
    - FLVN00 (FI Vendor)
    - FLVN01 (Vendor)



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Vendor Master Views

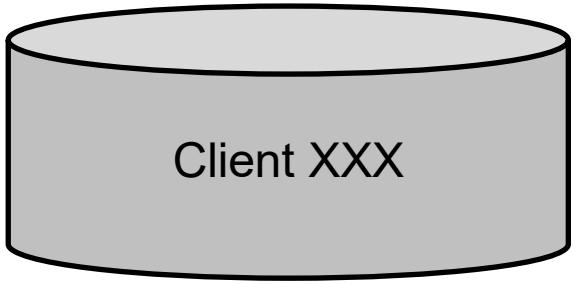
- Client Level
  - Address
  - Vendor Number
  - Preferred Communication
- Company Code Data
  - Reconciliation Account
  - Terms of Payment
  - Bank Account
- Purchase Org Data
  - Purchasing Currency
  - Salesman’s Name
  - Vendor Partners



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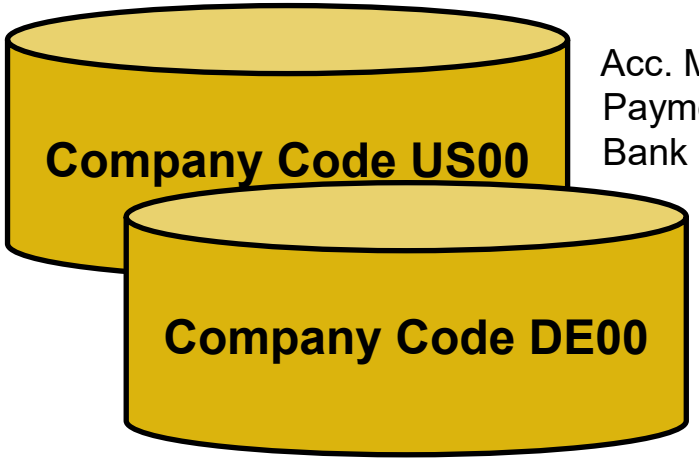
Vendor Master

General Information relevant for the entire organization:



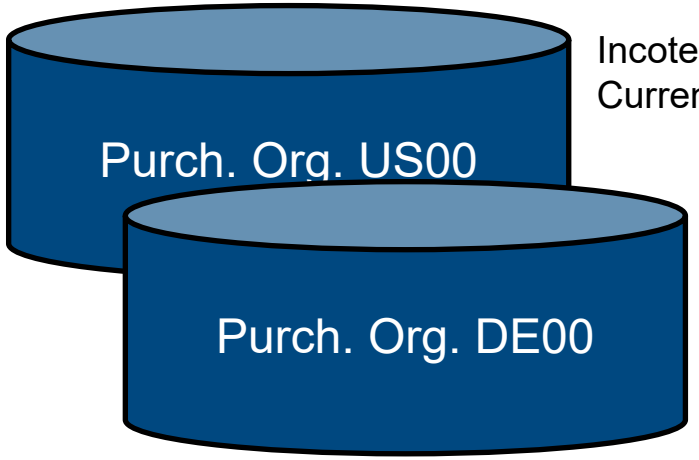
Name
Address
Communication

Company Code specific information:



Acc. Mgmt
Payment
Bank

Purch. Organization specific information:

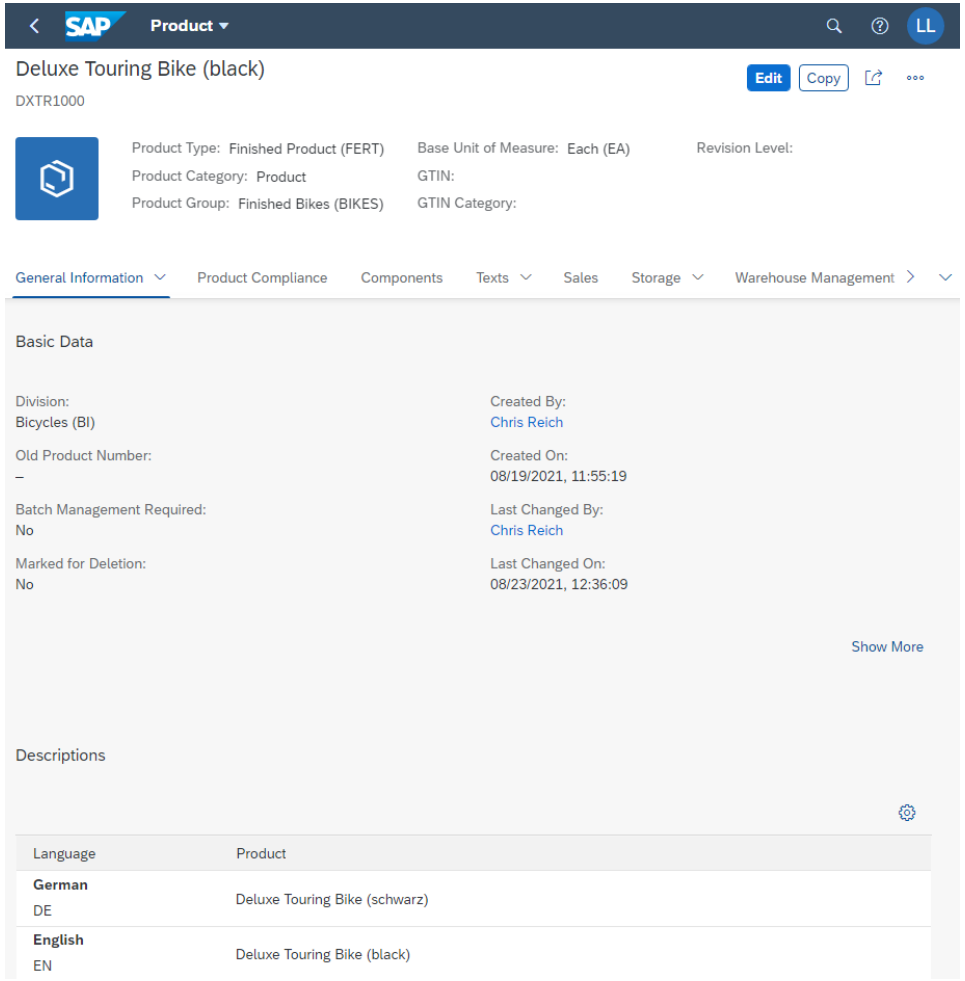


Incoterms
Currency

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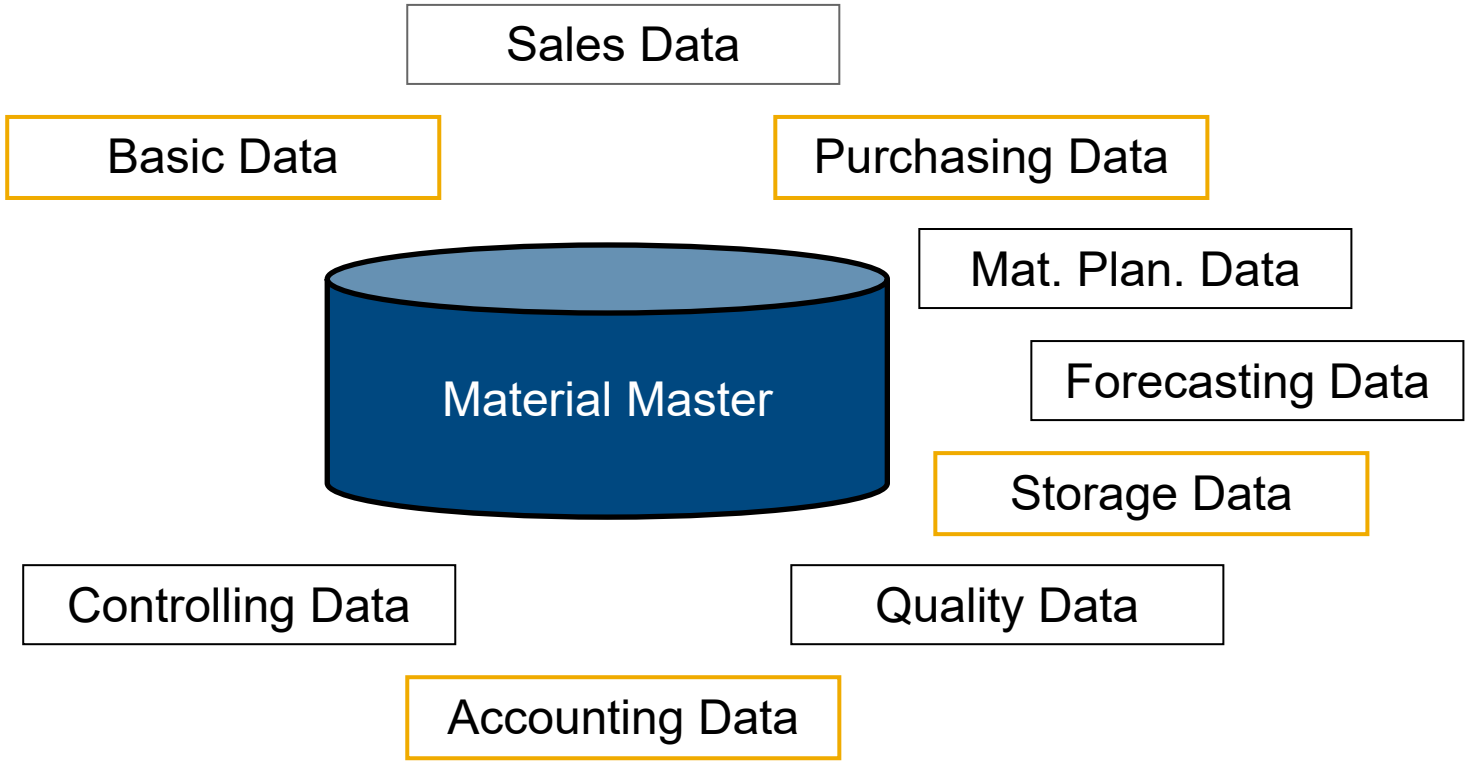
Material Master Data

- Material Master
  - Contains all the information a company needs **to manage material**
  - It is used by **most components within the SAP** system
    - Sales and Distribution
    - Materials Management
    - Production
    - Plant Maintenance
    - Accounting/Controlling
    - Quality Management
  - Material master data is stored in **functional segments called Views**



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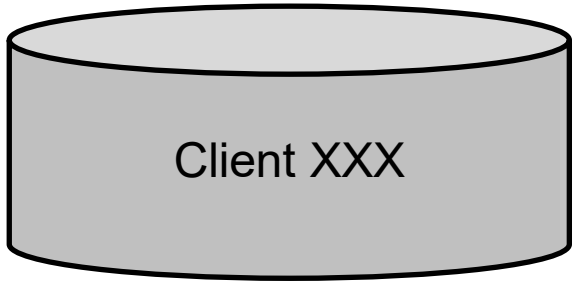
Material Master Views



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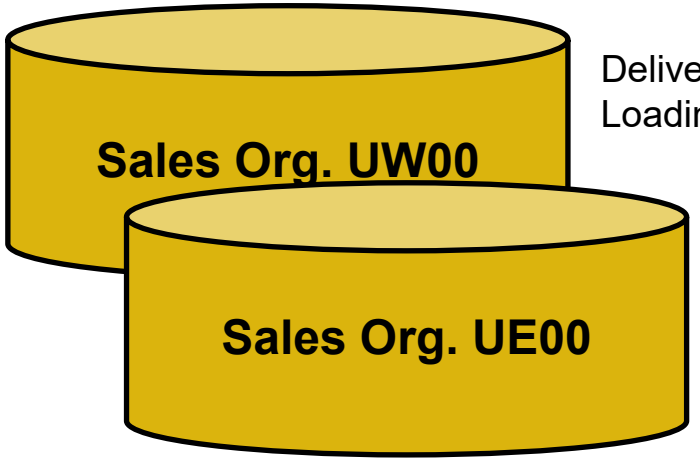
Material Master

General Information relevant for the entire organization:



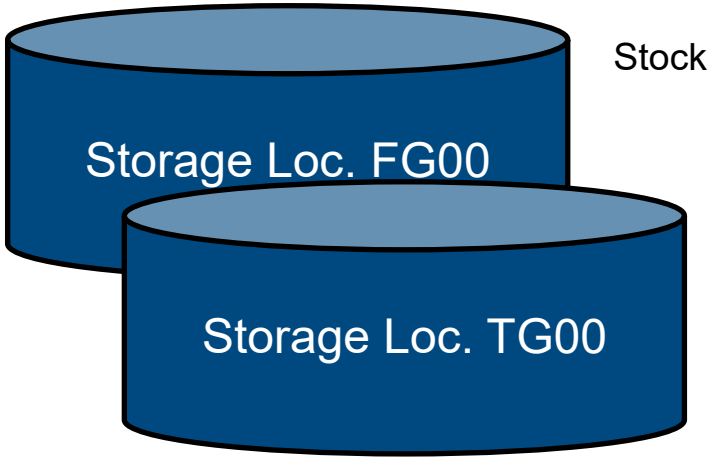
Name
Weight
Unit of Measure

Sales specific information:



Delivering Plant
Loading Grp

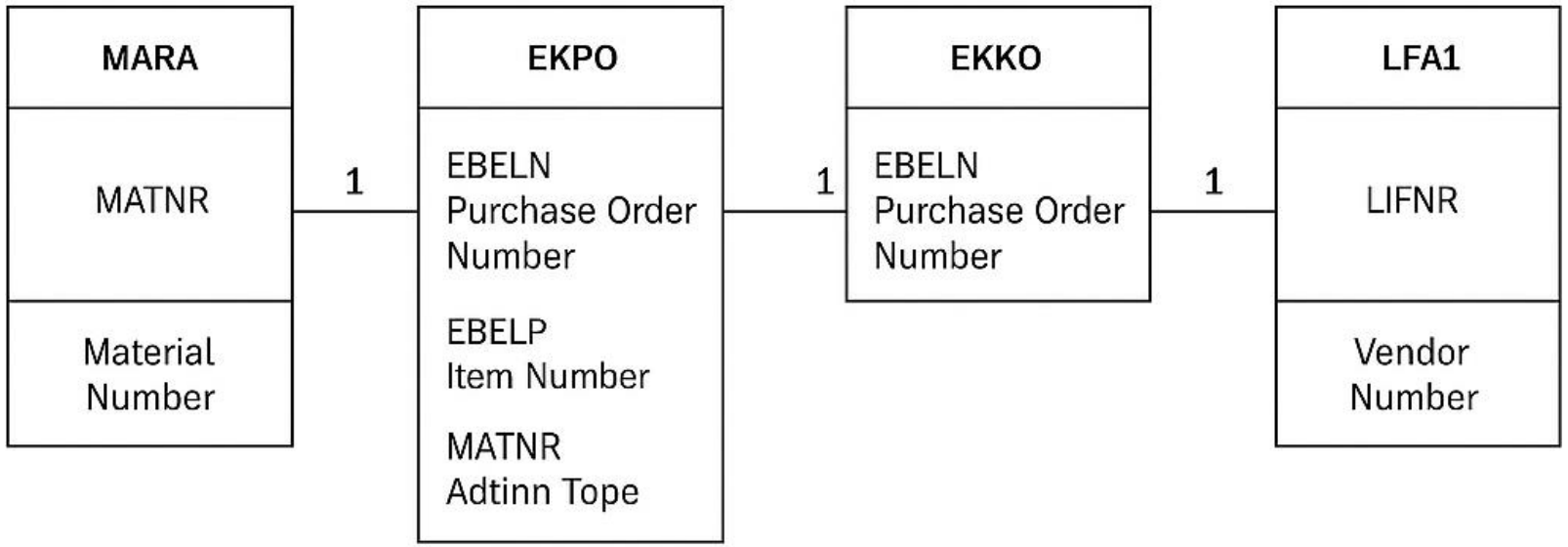
Storage Location specific information:



Stock Qty

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Data Model: Purchase Order (Summary)

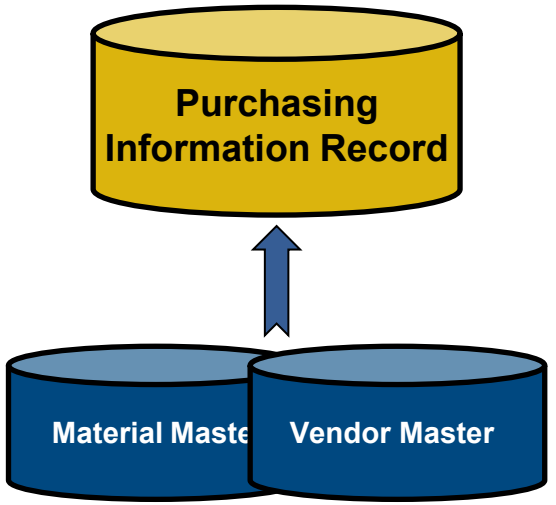


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Purchasing Information Record

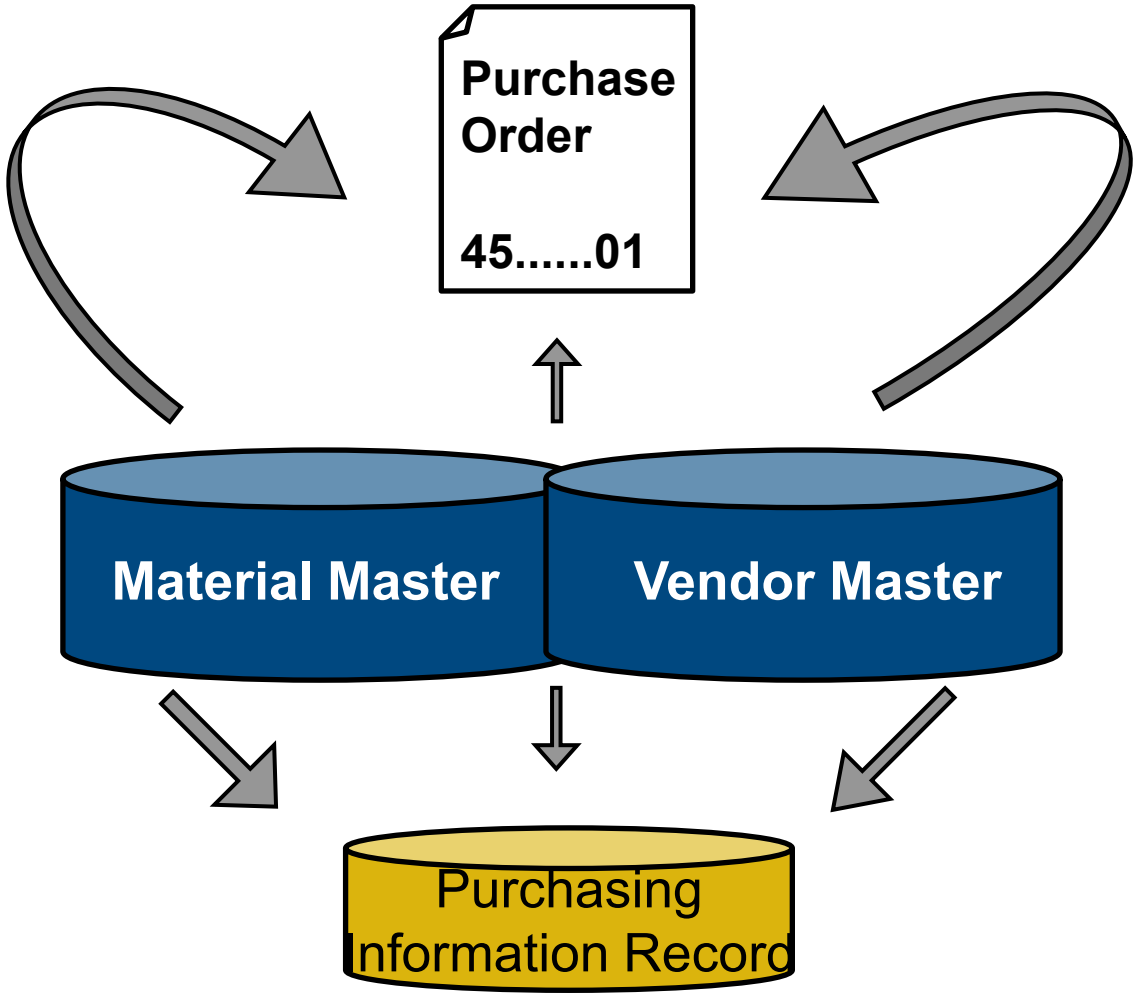
- Framework for Purchase Order
  - Contains the **relationship between a vendor and a material**
- Can be created:
  - Manually
  - Automatically – Quotations
  - Automatically – Purch. Orders
- **Reporting**
  - **Vendor Evaluation**

- Allows buyers to quickly determine:
  - Which **vendors have offered or supplied specific materials**
- Info Records contain:
  - **Data on pricing and conditions**
  - **Last purchase order**
  - Tolerance limits for deliveries
  - Specific lead times
  - Availability periods
  - Vendor Evaluation data
- Serves as default information for Purchase Orders



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Master Data in Use



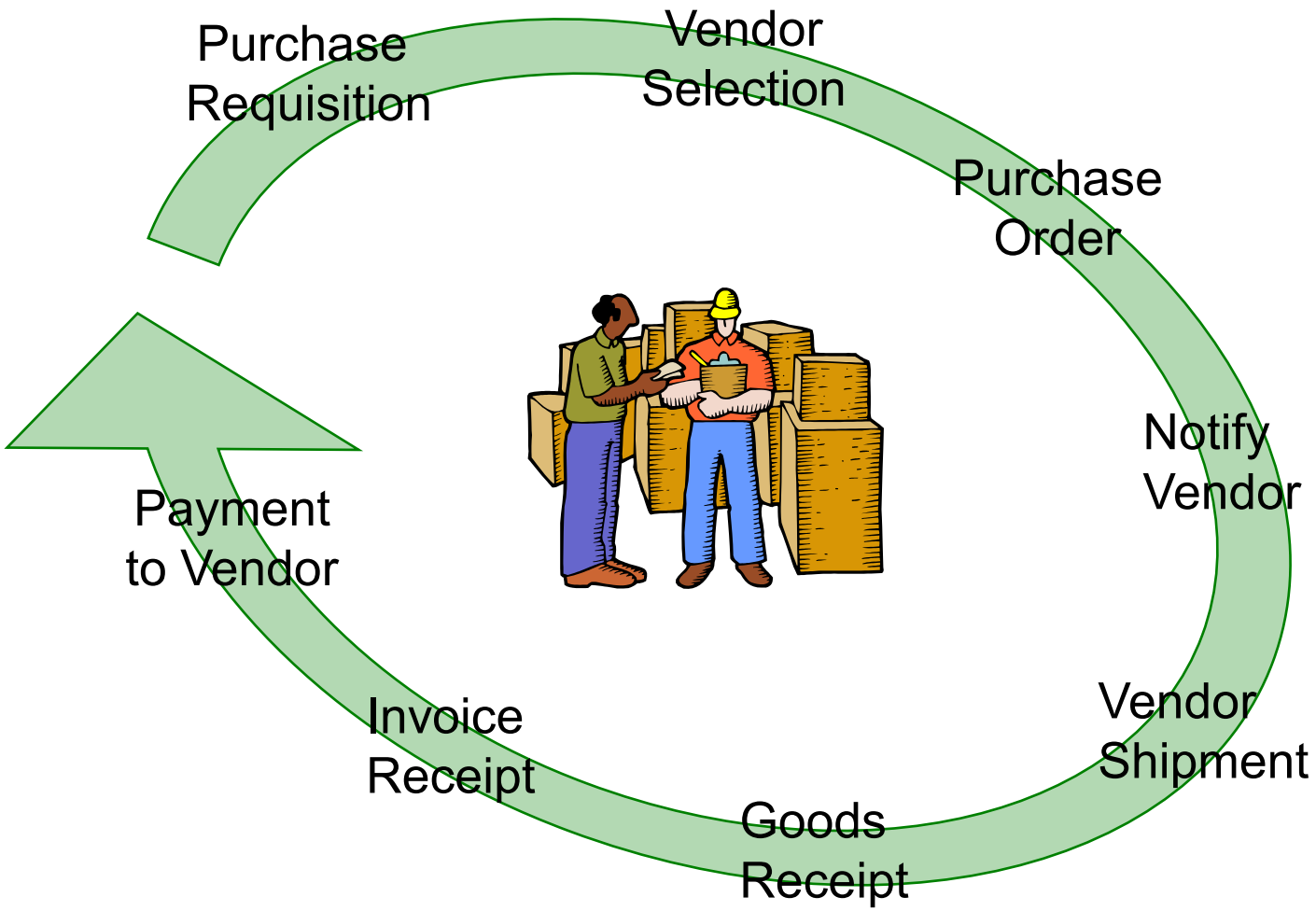
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Unit Overview

- MM Organizational Structure
- MM Master Data
- Procure-to-Pay Process

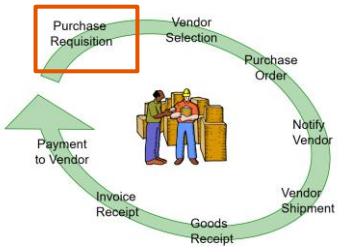
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Procure-To-Pay Process



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Purchase Requisition



- Internal Document **instructing the purchasing department to request a specific good or service** for a specified time

- Requisitions can be created two ways:
  - Directly - Manually
    - person creating determines: what, how much, and when
  - Indirectly - Automatically
    - MRP
    - Production Orders
    - Maintenance Orders
    - Sales Orders

- Once the requisition has **been assigned a source of supply it can be released for processing**

- There are a variety of ways that a purchasing department can process a requisition to determine the appropriate Source of Supply:
  - Internal Sourcing Requirements
  - Source List
  - Outlined Agreement
  - RFQ

➔ see following slides

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Source of Supply: Internal Sourcing and Source List (1)



Internal Sourcing

- The requisition for materials could be **satisfied by sources within our company**.
  - It is possible that a plant within your firm could represent a potential source of supply for the material needed (centralized warehouse)
  - If an internal source is identified the requirement is **covered by an internal procurement transaction (stock transport order)**

Source List

- A source list is a r**ecord that specifies the allowed means for procuring a material for a certain plant** within a given time period.
  - If the list contains a sole source the system will assign the vendor to the requisition.
  - If several options exist the system will **display a list of vendors** for you to choose from.
  - If no source has been established the system will revert to search information records and outline agreements.

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Source of Supply: Outline Agreement and RFQ

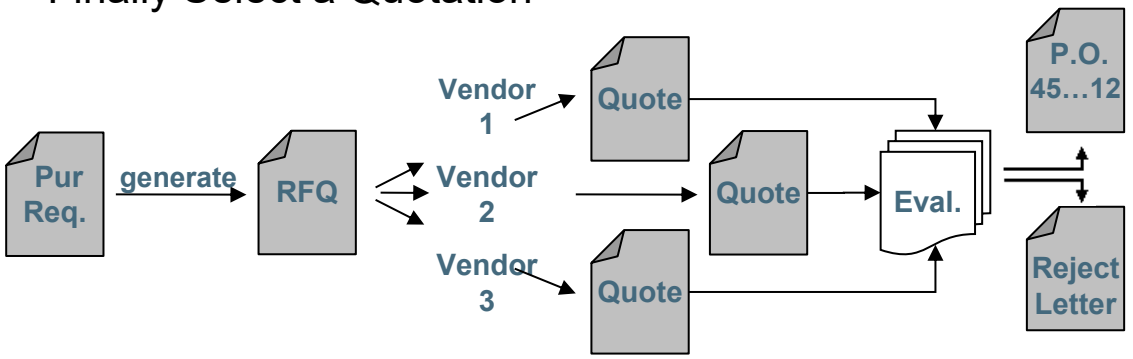


Outline Agreement

- Requisitions can be satisfied **through existing longer-term purchasing agreements**
- These agreements are subdivided into:
  - Contracts
    - Consists of items defining the individual materials, material groups, or services with prices and in many cases quantities
      - Quantity
      - Value
  - Scheduling Agreements
    - Total quantity of material is spread over a certain period in a delivery schedule, consisting of line items indicating quantities and their planned delivery date

Request for Quotation (RFQ)

- If nothing exist in the system we may need to submit a request for quotation to our vendors. An **RFQ is an invitation to a vendor by a Purchasing Organization to submit a bid for the supply of materials or services**
  - The accepted quotations will generate Purchasing Information Records
  - Perform Quotation Price Comparisons
  - Finally Select a Quotation



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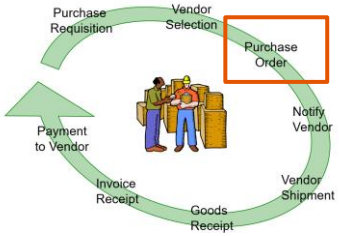
Vendor Evaluation once Identified



- Vendor evaluation **helps purchasing evaluate vendors for sourcing while also enabling the company to monitor vendor relationships through performance scores and criteria** you put in place.
  - Supports a maximum of 99 main criteria and 20 sub criteria for each main:
    - Price
      - Price Level/Price History
    - Quality
      - Goods Receipt
      - Quality Audit
      - Complaints/Rejection level
    - Delivery
      - On-time delivery performance
      - Quantity reliability
      - Compliance with shipping instructions
      - Confirmation Date
  - You then must establish a scoring range (1 -100) and determine the weight factors of scores for each.

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Purchase Order



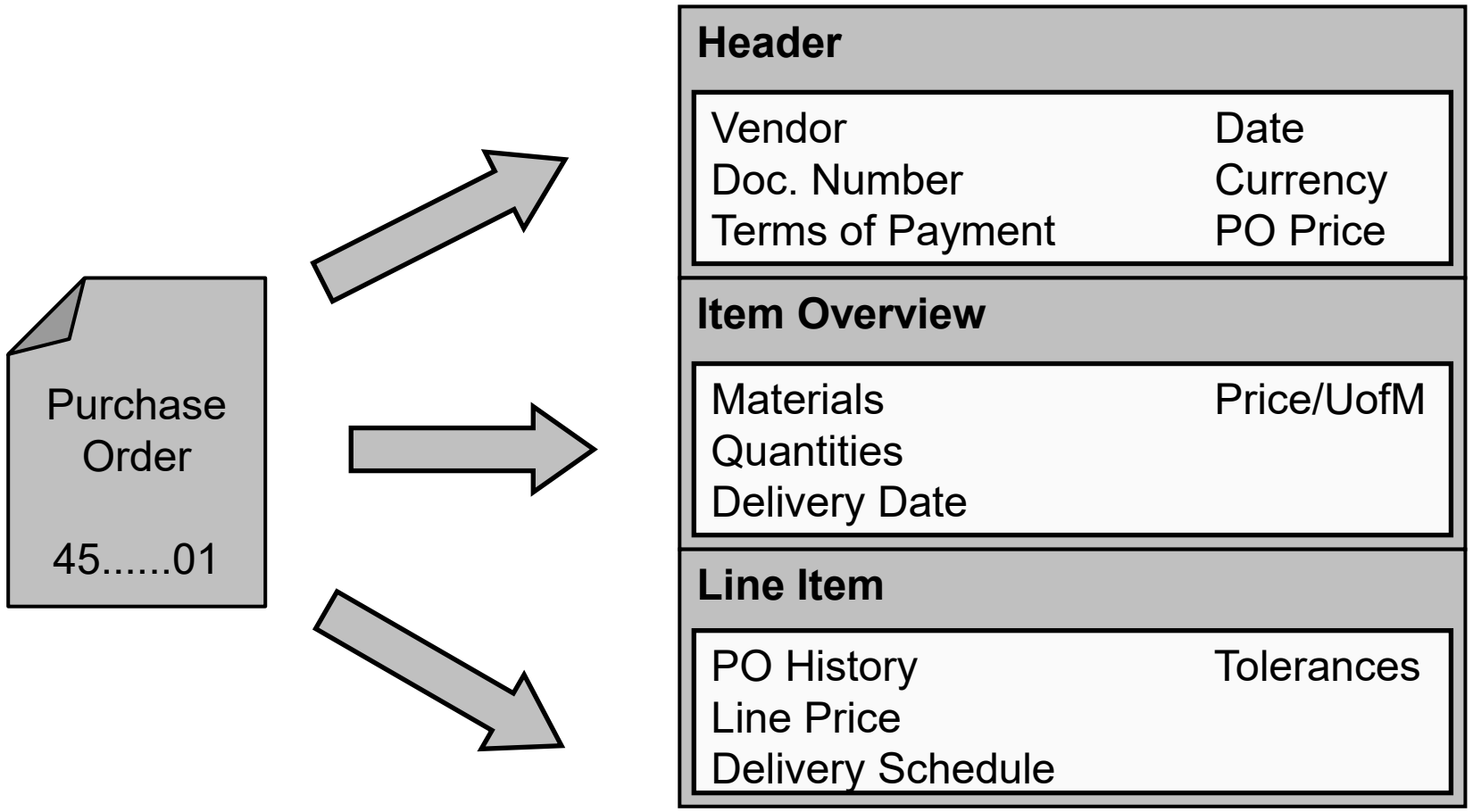
- A purchase order is a **formal request to a vendor for a specific material or service under the stated conditions**

- Purchase Orders can be created manually
  - Reference a Purchase Order
  - Reference a Purchase Requisition
  - Reference a RFQ/Quotation
  - Without Reference

- Purchase Orders can be created automatically

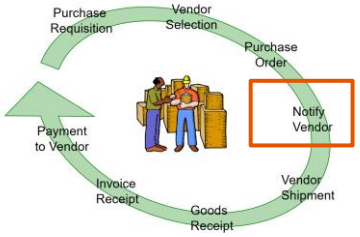
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Purchase Order Structure



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Purchase Order Output

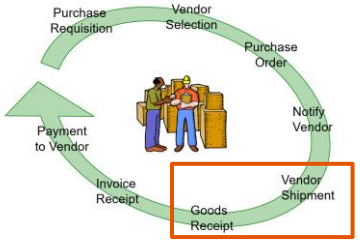


- Once a Purchase Order has been created the **vendor needs to be notified**
  - Printed
  - E-mail
  - EDI
  - Fax
  - XML

- There are a variety of forms that aid in the purchasing process and are generated from the Purchase Order
  - Purchase Order Output
  - Order Acknowledgement Forms
  - Reminders
  - Schedule Agreements

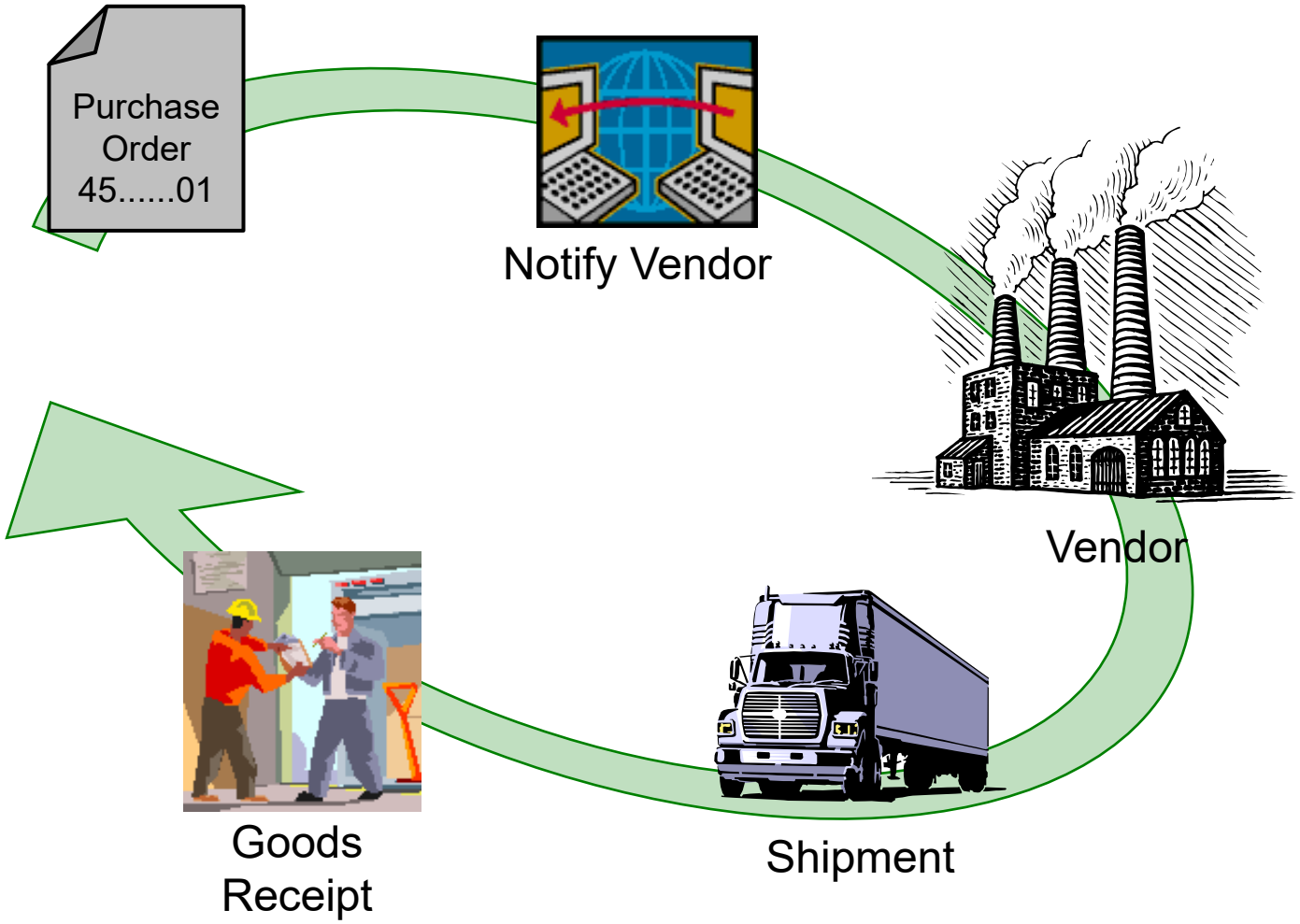
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Goods Receipt



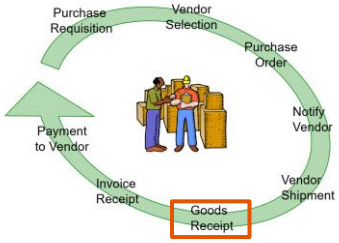
- Goods Receipt: Goods movement **in which we accept goods into our system**

- If materials are **delivered against a Purchase Order we will reference** that order to
  - Determine if we **got what we ordered**
  - System can process data for us from the PO
    - Material, quantity
  - Purchase Order History is updated with the receipt
  - **Updates Physical Inventory**
  - **Updates Inventory G/L Account**



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Material Movements



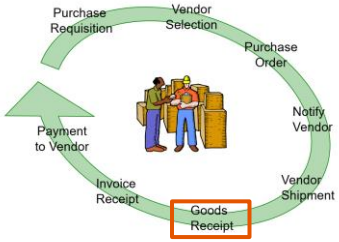
- When a goods movement takes place it is **represented by a Movement Type**

- Movement types are three-digit keys used to represent a movement of goods
  - 101 – goods receipt into warehouse
  - 103 – goods receipt into GR blocked stock
  - 122 – return delivery to vendor
  - 231 – consumption for a sales order
  - 561 – initial entry of stock

- Destinations for Receipt of Goods
  - Warehouse – Unrestricted, Quality, Blocked
  - Quality
  - Goods Receipt Blocked Stock

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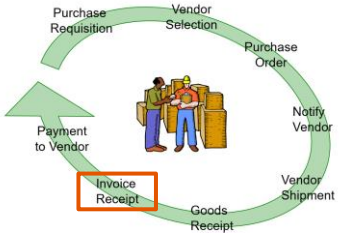
Effects of a Goods Receipt



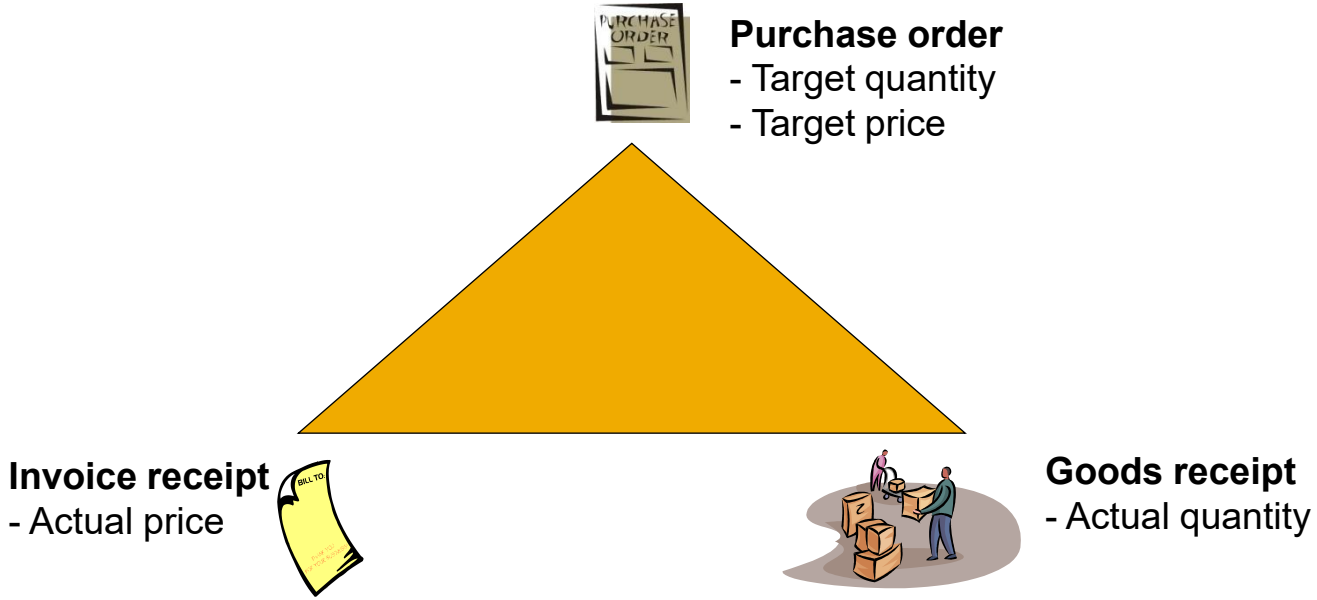
- When a Goods Movement for the receipt of goods takes place a series of events occur
  - Material Document is Created
  - Accounting Document is Created
  - Stock Quantities are Updated
  - Stock Values are Updated
  - Purchase Order is Updated
  - Output can be generated (GR slip / pallet label)

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Invoice Processing

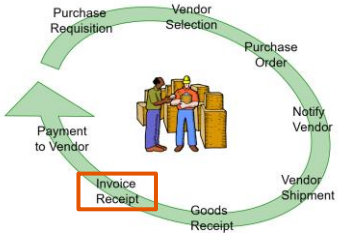


- Incoming Invoices are **referenced against a Purchase Order to verify their content, prices etc.**
- If discrepancies arise between the purchase order or goods receipt and the invoice the system **with generate a warning or an error**
  - Depending on system configuration the difference could cause the system to Block the Invoice



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Invoice Processing



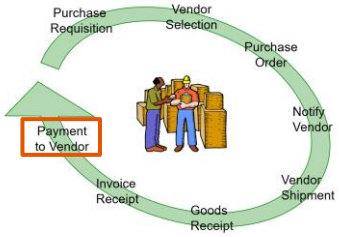
- When an invoice is saved **it creates a liability on the vendor's accounts payable account** with the value of the Goods Receipt of our Purchase Order

- Upon verification the:
  - Purchase Order is updated
  - Material Master is Updated (MAP)
  - **Accounting Document is created**

- Once the Invoice has been posted the verification process is completed and the payment process is initiated within Financial Accounting

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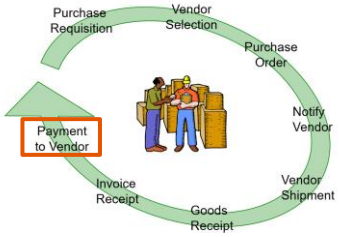
Payment to Vendor



- Can be done automatically or manually
- Elements of the Payment Transaction:
  - Payment Method
  - Bank from which they get paid
  - Items to be Paid
  - Calculate Payment Amount
  - Payment Medium
- Process will create a **financial accounting document to record the transaction**

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Goods Receipt / Invoice Receipt Reconciliation Account



Purchase requisition

Purchase order

Goods receipt

No impact on Financial Accounting (FI)

Materials Management (MM) and Financial Accounting (FI) via automatic account assignment

Inventory

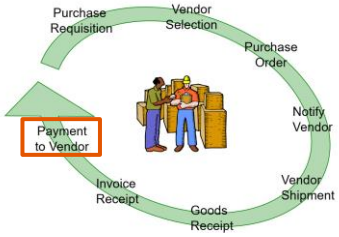
Dr	Cr
$100	

GR / IR

Dr	Cr
	$100

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Goods Receipt / *Invoice Receipt* Reconciliation Account



Invoice receipt → Amount owed is assigned and transferred to vendor account payable

GR / IR

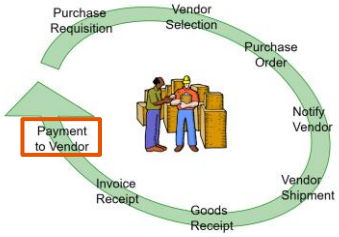
Dr	Cr
$100	

Vendor A/P

Dr	Cr
	$100

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Vendor Payment



Bank → Amount owed is paid to vendor and account payable is reduced

Bank

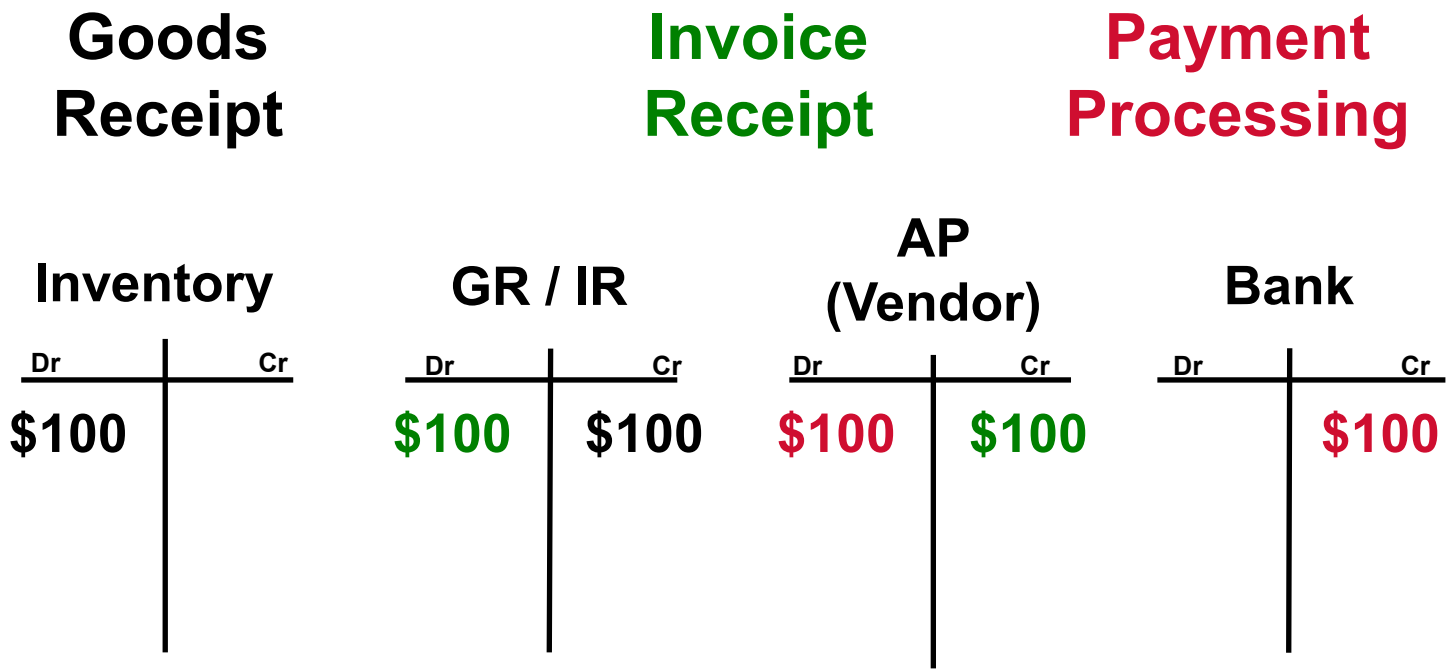
Dr	Cr
	$100

Vendor A/P

Dr	Cr
$100	

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FI – MM Integration Point



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Unit Overview

- MM Organizational Structure
- MM Master Data
- Procure-to-Pay Process

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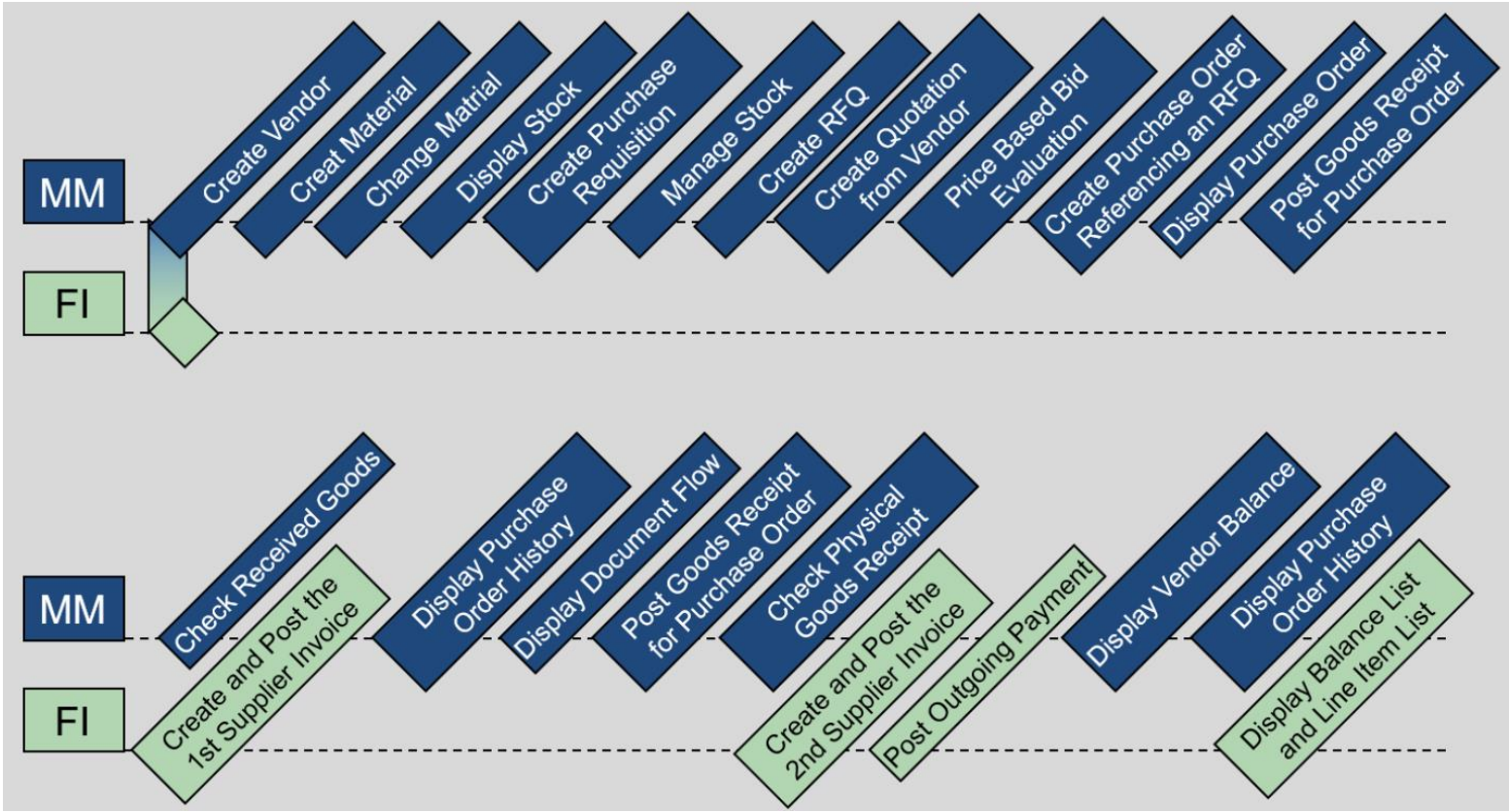
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Agenda

- Purchase-to-pay vs. Order-to-cash processes
- MM Organizational Structure
- MM Master Data
- Procure-to-Pay Process
- Outlook Case Study MM

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Case Study MM



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Learning Objectives (What?)

Students can…

- … describe the **general Procure-to-Pay Process on the basis of the essential activities** and **differentiate it from an Order-to-Cash process**.
- … describe the **organizational structures and master data** required for Materials Management with SAP SD.
- … create **vendor and material master data relevant for Materials Management** with SAP-MM.
- … create purchase requisition, purchase order and post a goods receipt in SAP-MM.
- … **explain the integrated order-to-cash process and execute** it in SAP-MM.